| □Backend Engineer

010-9101-5429 | digle117@gmail.com | github.com/PreAgile | astro-paper-23v.pages.dev

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□□ □□□□ 6□ □□ □□□□ □□□□ □□ SaaS □ □□□□ □□□□ □□□, □□□ **API □□ 20□ · □□□ □□□□ 100□ · □□ □□ 12□** □ □□ □□ □□□ □□□ □□·□□·□□□ □□□□□. **Kotest organization** □□ □□(6 PR □□), **LINE Armeria** (1 PR □□), **Spring Batch · Spring Cloud Gateway** □□ □□ □□ □.

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□□	**Before → After**	□□
□□□ □ □□	**□ 800□ → 90□ □ (88.75% ↓)**	□□ IP □□ □□□, □□ □□□ □□
□□ □□□ □□ □□□	**70% → 98%**	□□ □□□□ □□
□□ □□□ (□□ 6□□)	**99.2%**	□□ □ □□□□□ FIFO + lease TTL
□□ □□ □□ □□	**0□ / 6□□**	Idempotency-Key + □□ □ □□
□□ □□□□ webhook race	**□□ + □□**	「 □□ □ □□□ □□□□

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Production — □□ □□□□□ □□ □□ □□

- TypeScript · NestJS · Node.js · TypeORM
- MySQL / Aurora · PostgreSQL · Redis (Cluster · Lua · □□ □ · Token Bucket)
- RabbitMQ (Quorum Queue · DLX) · □□□ □ application-level retry
- Docker · Docker Compose · Traefik □□ □□□ · AWS ECS · □□□□ □□ □□
- Prometheus · Grafana · APM (span tagging) · Slack Webhook · JFR

JVM / Spring Re-design — □□□ □□□□□□ □□·□□□ □□

- Java · Kotlin · Spring Boot · Spring Batch · JPA · QueryDSL
- Redisson · Resilience4j (Circuit Breaker · Retry · Bulkhead · RateLimiter · TimeLimiter)
- Kafka (Outbox · CDC · Consumer Group) · ShedLock · OAuth2

Evidence — □□·□□□·□□ □□

- Testcontainers · JUnit 5 · Kotest · Pytest · Jest · nGrinder
- Kotest organization □□ · Armeria □□ · Spring Batch □□ □□

□□□□ □□

JVM □□□ □□□□□ □□ □□ — □□ **7** · □□ □□ **+ PR** □□ □□.

[Kotest](https://github.com/kotest/kotest) — Kotlin □□□□□ □□□ □□□□□ (4.7k stars)

2026.03 □□□□□ sksamuel □ □□ □□□ **organization** □□ □□, □ □□ 6 PR □□.

- □□ □□ Test Metadata Public API □□ □□ (+355/-16, 5 □□ 16h □ □□) — [issue #5103](https://github.com/kotest/kotest/issues/5103), [PR #5905](https://github.com/kotest/kotest/pull/5905)
- □□□ data class □□ □□ diff — [issue #2545](https://github.com/kotest/kotest/issues/2545), [PR #5835](https://github.com/kotest/kotest/pull/5835)
- JSON Schema anyOf / oneOf □□ □□ — [issue #4463](https://github.com/kotest/kotest/issues/4463), [PR #5807](https://github.com/kotest/kotest/pull/5807)
- JSON Matchers □□□ Json □□□□ □□ — [issue #4601](https://github.com/kotest/kotest/issues/4601), [PR #5795](https://github.com/kotest/kotest/pull/5795)
- @OnlyInputTypes □□ □□ □□ □□□ — [issue #5589](https://github.com/kotest/kotest/issues/5589), [PR #5789](https://github.com/kotest/kotest/pull/5789)
- shouldHaveSingleElement □□□ □□□ — [issue #5755](https://github.com/kotest/kotest/issues/5755), [PR #5756](https://github.com/kotest/kotest/pull/5756)

[Armeria](https://github.com/line/armeria) — LINE □ Java □□ □□□ RPC □□□□□ (5.1k stars)

- RequestContextExporter BuiltInProperty □□ + Logback MDC export □□□ □□ (+339/-5, **MERGED** 2026.04) — [issue #4403](https://github.com/line/armeria/issues/4403), [PR #6683](https://github.com/line/armeria/pull/6683)

[Spring Batch](https://github.com/spring-projects/spring-batch) — Spring □ □□ □□ □□ □□□□□ (2.9k stars)

□□ □□ □□, PR □□ □.

- fault-tolerant step chunk scanning □□ □□ 4□ □□ □□ — [issue #3946](https://github.com/spring-projects/spring-batch/issues/3946) (□□□□□ "documentation issue" □□ □□)
- conditional flow decider order □□ — [issue #4478](https://github.com/spring-projects/spring-batch/issues/4478) (□□□□□ "this is a bug" □□)

[Spring Cloud Gateway](https://github.com/spring-cloud/spring-cloud-gateway) — Spring □ reactive API gateway

□□ □□ □□, 5.1.x □□□ □□□ □□.

- Apache HttpClient5 □□ □□□ □□ □□□□ — [issue #3311](https://github.com/spring-cloud/spring-cloud-gateway/issues/3311) (□□□□□ □□□ + □□ □□)
- API Versioning Predicates □□ □□ □□ — [issue #4024](https://github.com/spring-cloud/spring-cloud-gateway/issues/4024) (□□ □□□ □□□□□ □□)

□□ — □□(Lemong) | | □ 2024.11 ~ □□

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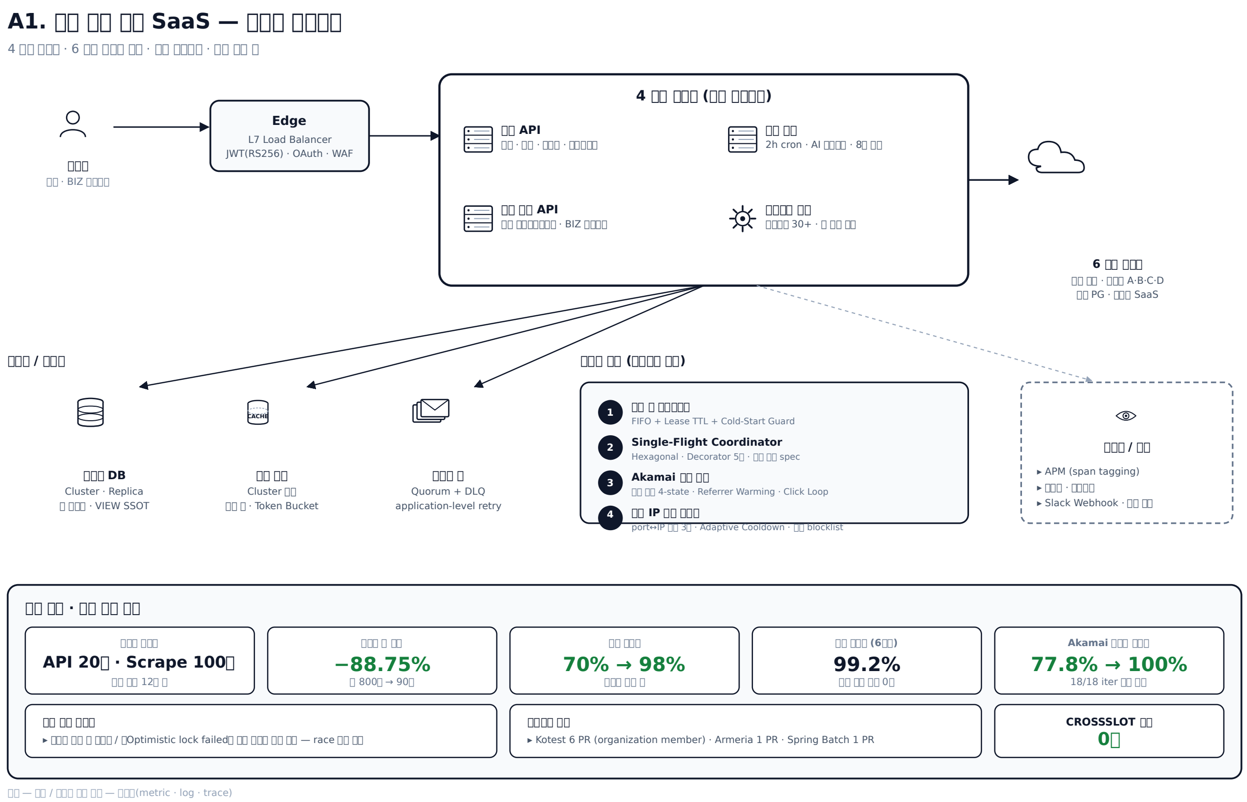

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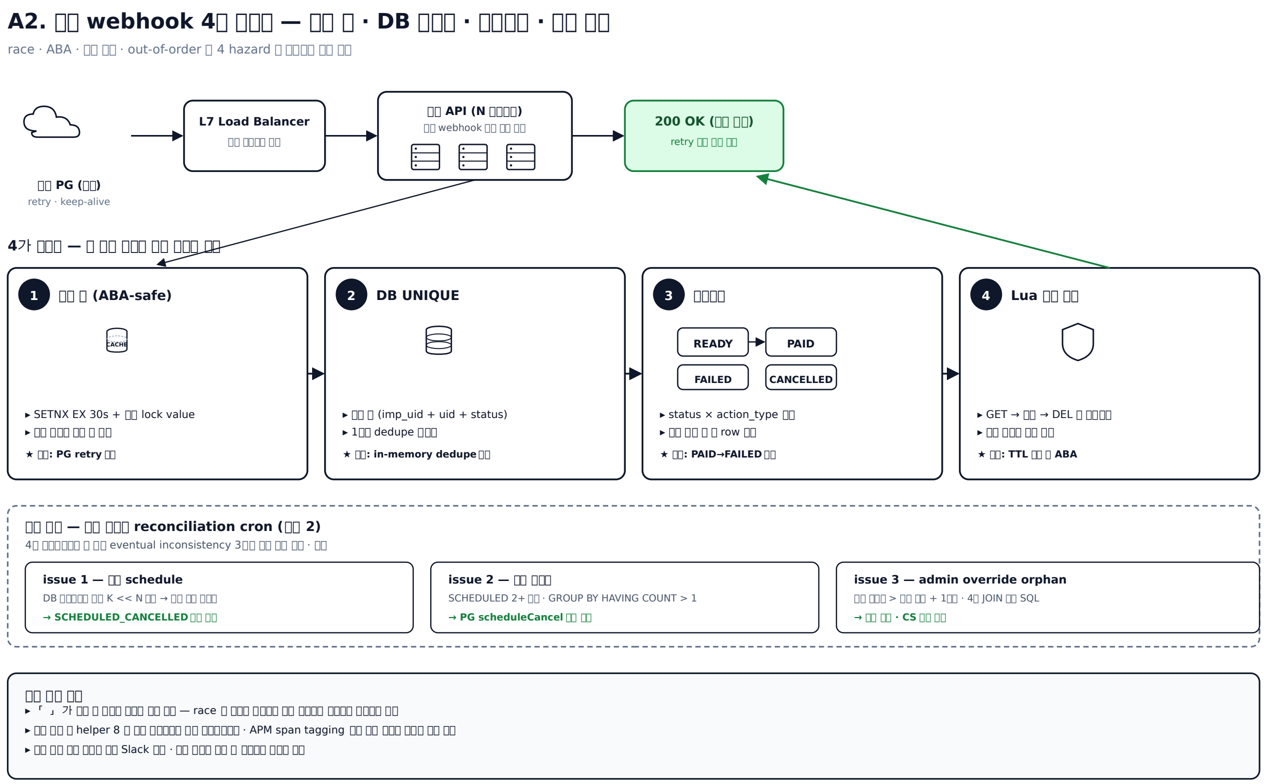
□□□ 5 □ □□ □□ + □□ □□ □□ □ □ □□ + □□ JVM/Spring □□□ □□□□ □□□□ □□□□□.

□□ □□□ + □□□ reconciliation — 「 」 □ □ □□ □□□ □□□□



□□ webhook 4□ □□□ + □□□ reconciliation

□□ — □□ PG webhook □ □□ □□ □□□□ □□ □□□□ □□ **5□**.

- PG □□ retry (5xx / timeout)
- □□□□ schedule 1□ □□ □ fallback retry → PAID / FAILED □□ □□ □□
- □□ □□ □ □□□ · □□ □□ □□ □□
- Load Balancer keep-alive □□□

□□ INSERT □ □□ row □□ □□ → □□ □□ → □□□ 2□ □□ → □□□□ □□ □□ → □□ □ □□ □□ □□□. □□ □□□□ □□□ in-memory dedupe 가, ORM □ findOne → save □□□ read-modify-write □□□ race window, □□ □□ □ □□□□ □ TTL □□ □ ABA □□ (Martin Kleppmann Redlock □□□□ □□), DB UNIQUE □□□□□ PG retry □□□ 1□ □□ □□.

가 webhook 4□ □□□□□ □ □□ □□ □□□ □□□ □□□□□ — □□ PG □ DB □ source of truth □ □□□□ □□ □□ schedule (PG □□□□ + DB cancel), □□ □□□, admin override orphan □□□□ □□□ □□□ / □□ □□ □□ □□□ □□□□□.

□□ — 「□ / DB / □□□□ / □□ □□□ 4□ □□ + □□ reconciliation cron □ □□ □□.

4□ □□□ (□□ □□):

1. □□ □ **(ABA-safe)** — □□ □□ SETNX EX 30s + □□ lock value. Stripe Idempotency Key □□□□ □□□□ lock value □ ownership □□

2. **DB UNIQUE** — □□ □ imp_uid + merchant_uid + status · 1□□ dedupe window · □□ fail-open □□□ □□ □□

3. □□□□ — READY → PAID / FAILED / CANCELLED × action_type □□ · □□ □□ (PAID □ FAILED □□) □ □□ row □ □□

4. **Lua** □□ □□ — GET → □□ → DEL □ □□□□ · □□ □□□ □□ □□, TTL □□ □ ABA □□

□□ reconciliation cron — 3□ inconsistency □□ □□:

- □□ **schedule** — DB □□□□□ □□ K << N □□ □ □□ PG getSchedulesByBillingKey □□ → □□ 4-way □□ □□ □□
- □□ □□□ — GROUP BY user_id HAVING COUNT > 1 □□ SQL
- **admin override orphan** — active.end_date > scheduled.schedule_at + 1month □□ 4□ JOIN SQL

□□ □□□□ □□ □□ — □□ □□ (findOne((user, coupon)) idempotent return) · □□□□ (QueryRunner □□□) · schedule □□ □□ amountWithUserCoupons fallback □□.

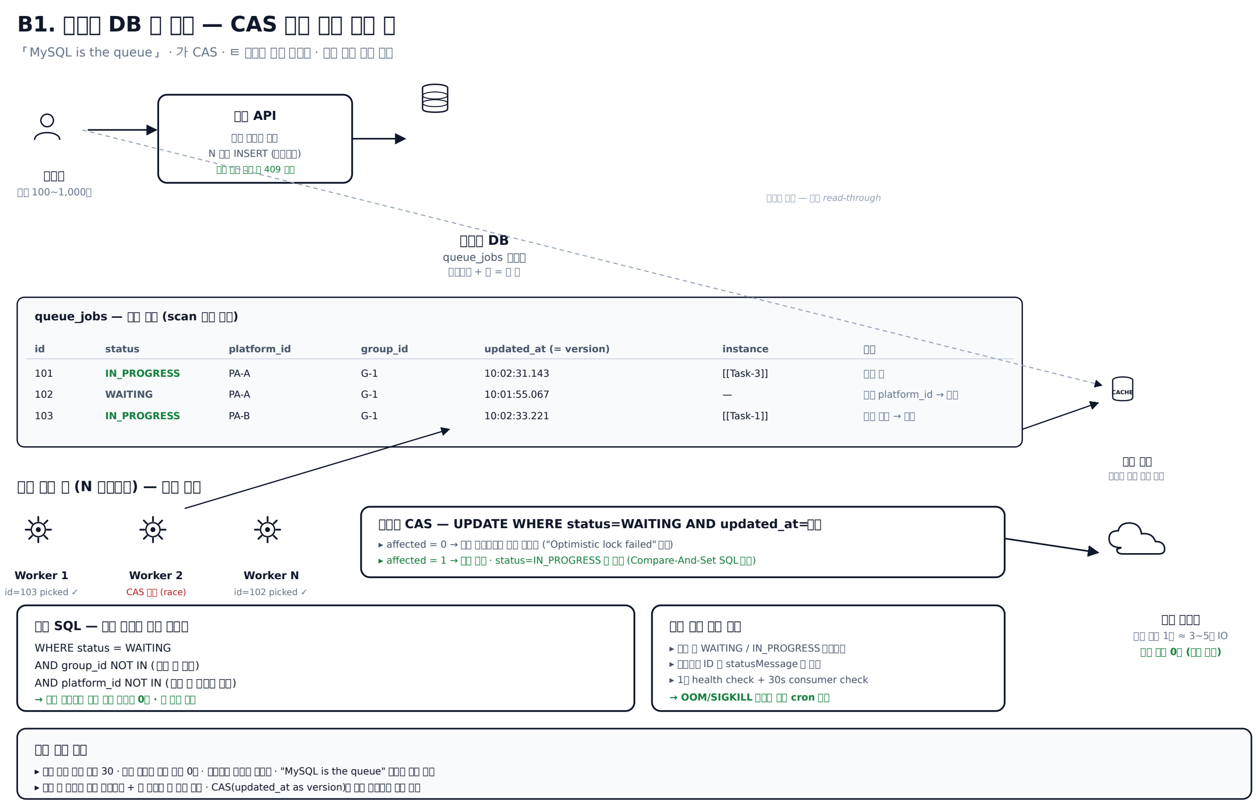
□□ (□□ □□)

- 「 □ □□□□□ □ □□: merchant_uid=...」 □□□ □□ □□ — race □□ + □□□□ □□□□ □□
- □□ SETNX + Lua DEL helper □ □□ □ 8 □ □□ □□ (promotional upgrade · change plan · verify code · □□ □□) □□ □□ helper □ □□□
- □□ 09:10 KST □□□ □□□ Slack □ □□□□ □□□□□ □□ — □□ □□ / CS □□□ □□ □□□ □□
- □□ □□ □□ 0□ (UNIQUE + findOne □□ □□), □□ □□ □ □□ update □□ 0□

□□ □□ — 4□ □□□ + reconciliation cron □□ · □□ · □□ □□□□ (Slack □□ □□ □□ □□) □ □□ □□ □□.

Spring □□ — @Transactional propagation · Redisson watchdog · JPA @Version OptimisticLockException · ER_DUP_ENTRY → @RestControllerAdvice · ShedLock cron · QueryDSL □□□ (□□□ □□□□ [commerce-comment-platform-be](https://github.com/PreAgile/commerce-comment-platform-be) □□ □□ □□□□ □□)

□□ □□ □□ — DB □ □□ □□ CAS + RabbitMQ □□ □□



□□□ DB □□ □ + Hierarchical □□□

□□ — □□ □□□ □□□ □ □□ □□□ □□□□□.

- **□□ □□□ □□ □□ (□ □ ≈ 3~5□ IO)** — □□□□ 「100~1,000」 □□□ □□□. HTTP □□ □□ □ Load Balancer 5□ □□□□ □□, □□ □□□□ in-memory queue □ □□□ □ □□ □□, □□ □□□ □□ □□□ □□□□ worker 2□□ □□ □□□ □ □□ □□□ □ □□ → □ □□ □□ □□
- **2□□ cron □□ □□** — □□ □□□ 1□ □□ □□ **13,127 □□ / 5,719 □□ □□ / 13,699 task** □□. □□ □□□□ □□ □□ □ □ □□ □□ □ captcha / □□□□ / □□ □□□. □□ □□ ThreadPool □ □□ □□ □□□ □□ □□ □□ □□□ = □□ □□, □□ □□□ 11□□ → 2□□ cron □□ □ □□

□□ □□□ □□ fire-and-forget □ □□□□ stateful □□·□□·□□·□□□ □□□ □□□. BullMQ □□ □□ □ □□□□□ □□□□□ □□ □□ □.

□□ — □ □□□ □□□□□.

(a) **□□□ DB □ □□ □□ (CAS □□)** — □□□□□ □ □□□ □ □□ □□. □□ □□ □□ □□□ updated_at □ version □□□ □□ ORM □□□ = CAS (Compare-And-Set) SQL □□. affected = 0 □□ □ □□□. □□□□□ □□ □□ □□□ □□ SQL □ platform_id NOT IN (□□ □ □□) □□□□ □□.

(b) **□□ □□ □□ / □□ □□ □□ □ hierarchical □□□** — (platform_id, password) □□ □□ □□□ → ThreadPool □ □ □ □□ □□. □ □□□ □ worker thread □ □□□□ □□□ □□ □□. □ □□ □□ □□ □ _deactivate_shops □ □□ □□ □□ □□ (□□ □□ □□) — □ □□ □□ □□.

□□ (□□ □□ □) — □□ □□□□ □□□□ □□□ □□ □□□ AAS □□ □□□ □□ → **RabbitMQ** □ □□□□ □□□□□□ □□ □. □□ □□ / □□ □stateful □□□ fire-and-forget □□□ □□ □□□ □□□, DB □ □□ □□□ source of truth □ □□ □□ fan-out □ □ □□□□ □□ □□□.

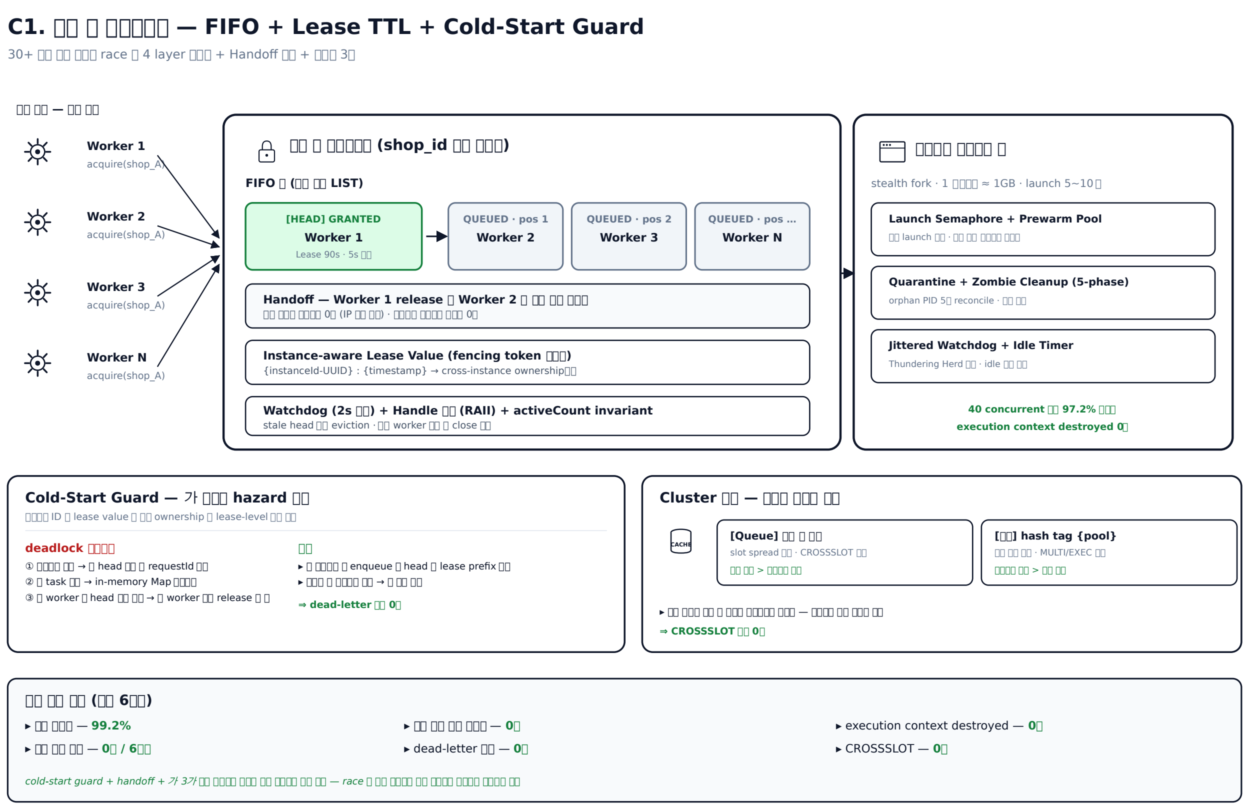
□□ (□□ □□)

- □□ □□ □□ □□ 30, □□ □□ □ □□ + □□ □□ □□□ □□□ □□□
- □□ □□ Job N modified by another instance (optimistic lock failed) □□ □□ — race □□□□ □□□□ □□□□ □□
- □□ □□□ □□ □□ 0□, □□□□ □□□ □□□
- □□ □□□ 13,127 □□ 2□□ cron □□ □□, □□ platform_id □□ □□ 0□
- □□ □□ □□ □□□ □□□ · □□ □□ □□ □□ (□□ □□)

□□ □□ — RDS Queue □□ / □□ SQL CAS □□ / □□ □□ □□ □□ (□□ □ □□□□ + health check) □□ □□. hierarchical □□□ □□ + □□ □□ □□□ dict □□. □□ RabbitMQ □□ □□□□□□ ADR □□ □.

Spring □□ — JPA @Version □□□ + OptimisticLockException (CAS) · Spring Batch + ShedLock + QuerydslZeroOffset Reader · Resilience4j Bulkhead.semaphore (□□ □□ □□□) + CircuitBreaker (□□□ □□) · Kafka Outbox / Consumer □□ + DLQ (□□□ □□□□ [commerce-batch-orchestrator](https://github.com/PreAgile/commerce-batch-orchestrator) □□ RabbitMQ↔Kafka □□ □□)

□□ □□□ □□ + □□ □□□ — 30+ □□ □□ □□□ race · □□ □□□ □□□ □□



□□ □ □□□□□ + Cold-Start Guard

□□ — 30+ worker □ □□ □□ □□□ □□□ □□□ □□□ □□ race + □□ □□□□ □ □□ □□□□ □□□□ □□□□□ □□□□ □ □□ □□ □ □□ □□□ □□□□ □□□□.

□□ race:

- □□ □□ □ page □□ navigation = □□□□ □□□□ execution context destroyed
- □□ □□ □□ □□□ □ □ = □□ □□□ IP □□ □□ □□ (1~24□□ □ □□ □□ □□ □□)
- □□□□ □□□□ 1 □□□□ ≈ 1GB+, launch 5~10□ — race □ □□□·□□□ N□ □□
- □□ □□ Cluster □□□ □□ □□□□ □□ □ □□ □ CROSSSLOT
- 가 □□□□ hazard: **cold-start deadlock** — 가 □□□ □□□□ in-memory Map □ □□□□□ □□ □□ □ head □ □ □□□□ requestId. □□□ worker □ □ head □□□ □□□□□ □ worker □ □□□ □□□ release □ □

□□ □□□ □□□:

- □□ □□ □□□□ □ □□ □□□□ □□□ □□□□ □□□□□ 99% □□
- □□ □□□□ □□ □□□ pending → verified → blocked □ □□ □□□ □□□, □ □□□ 4 □□ silent □□□□ □□ □□□ password retry □ □□ □□□□

□□ — □ □□ □□□□ □□□ □□□ □□□ □□ + □□ □□□ □□□ □□□ □□.

(a) □□ □ □□□□□ **— 4 layer** □□□:

- FIFO □ (□□ □□ LIST) · per-request lease (90s TTL · 5s □□)
- **Instance-aware lease value** = {instanceId-UUID} : {timestamp} (Martin Kleppmann fencing token □□ □□□ — cross-instance ownership □□□ lease-level □□)
- Handoff pattern (release □ □□ □ entry □ □□ □□ □□□)
- □□□ 3□ — forceRelease (□□ kill switch) · forceTerminate (activeCount □□ 0) · evictStaleHead (2s □□ stale head LREM)
- □□ □□ Cluster CROSSSLOT — queue □ □□ □ □□ (□□ □□ □□), □□ □□□ hash tag {pool} (□□□□ □□ □□) □□ □□□ □□□ □□ □□

(b) □□ □□□ □□□ □□ **—** □□ □□ □□□:

- □□ □□ 4-state □□□□ (□□ / □□ / challenge / □□) — 「 □ □□ > challenge > □□ > 」 □□□□ □ □□□ 4 □□ silent □□□ □□ □□
- □□ □□□ service □ spec □ SSOT □□ □□□ □□ (discriminated union □□)
- Referrer Warming — □□ URL □□ referrer chain □□, 15□ human-like mouse jitter □□ → sensor telemetry □□ → □□ □□ □□
- □□ □□ helper □ main loginResult / Quick Retry / Prewarm Swap 3 □□□□ □□ □□ (silent □□ □□)

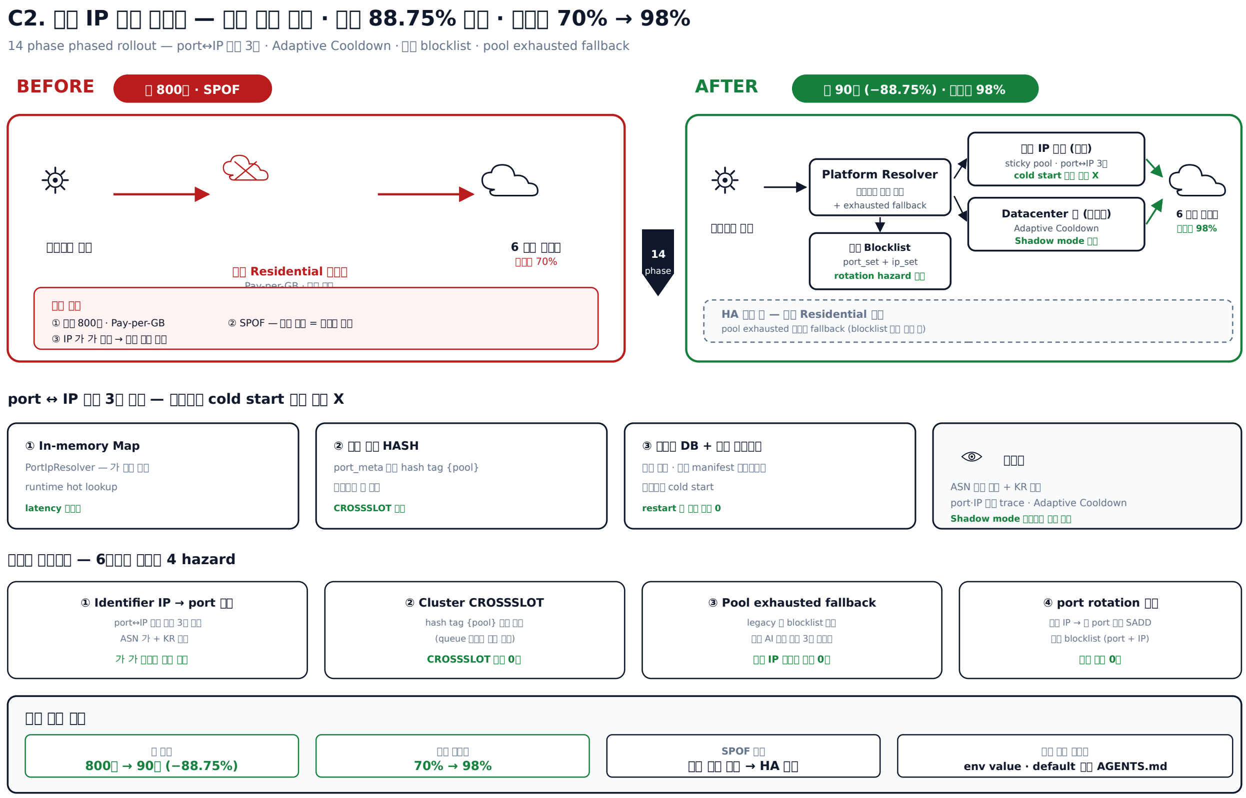
□□ (□□ □□ · □□ 6□□)

- □□ □□□ **99.2%**, □□ □□ □□ □□□ **0**□
- □□ □□ □□ **0**□, execution context destroyed □□ **0**□
- □□□□ □□□ □ dead-letter □□ **0**□ (cold-start guard □□)
- CROSSSLOT □□ **0**□ (hash tag / □□ □ □□ □□ □□ □□)
- □□ □□□ □□□ □□□ **77.8% → 100%** (Referrer Warming □□ · 18/18 iter □□ □□ · PR □□□ iter × worker □□□ □□ □□ □□□ □ □□□□ □□□□)

□□ □□ — □□ □ □□□□□ (4 layer + □□□ 3□) □□ · □□ · □□. □□ □□ □□□□ □□ □□ spec / service SSOT □□ □□□□.

Spring □□ — Redisson RFairLock (FIFO + lease auto-extend) · Kotlin Coroutines Mutex.withLock per-key in-process layer · sealed class □ □□ □□ □□ discriminated union · Resilience4j Bulkhead □□ □□□□ □ □□ (□□□ □□□□ [commerce-external-gateway-kt](https://github.com/PreAgile/commerce-external-gateway-kt) □□ 7 □□ □□ □□ spec □□ □□□)

□□ IP □□ □□□ — □□ □□ □□ + □□ 88.75% □□



□□ IP □□ □□□ Before / After

□□ — □□ Residential □□□ □□ □□□□ □□ · SPOF · □□ □□□ □ □□ □□□□ □□□□□.

- **□□ / SPOF** — □□ 800 □ □ · □□ □□ □□ = □□□ □□
- **Identifier □□□** — Datacenter □□□□ port □□ IP □ □ㄴ □□ □□ → □□ □□ □□ (IP) □ stable identifier (port) □ □□□
- **Cluster CROSSSLOT** — □□ □□□ · STREAM · blocklist □ □ MULTI/EXEC □ □□ □ □□ □□
- **port rotation □□** — □□ IP □ □□□ □□ port □ mapping □□ □ port □ □□ blocklist □ □□

□□ — 6□□ 14 phase phased rollout. □□ □□□ □ phase □□ → □□ → □□ □□ → □□ phase □ □□□ cutover.

□□ □□ 4□□:

- **port↔IP □□ 3□** □□ (In-memory Map + □□ □□ HASH + □□□ DB + □□ □□□□ □□□□□ manifest) — □□□ □ cold start □ □□ □□ □□ □□
- **□□ blocklist** — port_set + ip_set. allocator □ port □□ □ IP lookup → ip_set hit □ □ port □ □□ port_set □. port rotation □□ hazard □□

- **Adaptive Cooldown** — 5□ outcome (success / block / timeout / networkError / authError / siteChange / unknownError) × consecutiveFailures × latency □□□□ health □□, Shadow mode □ □ □□ □□ □□ □□ □□
- **Pool exhausted → legacy fallback** — NaverIspPoolExhaustedError throw → legacy path □□, legacy □ selectPortRespectingBlocklist □ □□ IP skip

□□ □□□ AGENTS.md □ □□□ — env value □ □ · □□□ · deprecated alias □□ □□ □□ default □□ □□. fallback □□ □□□ □□ **AI □□ □□ 3□ follow-up** □□ □□□□ □□□ PR □□□ □□□□□□.

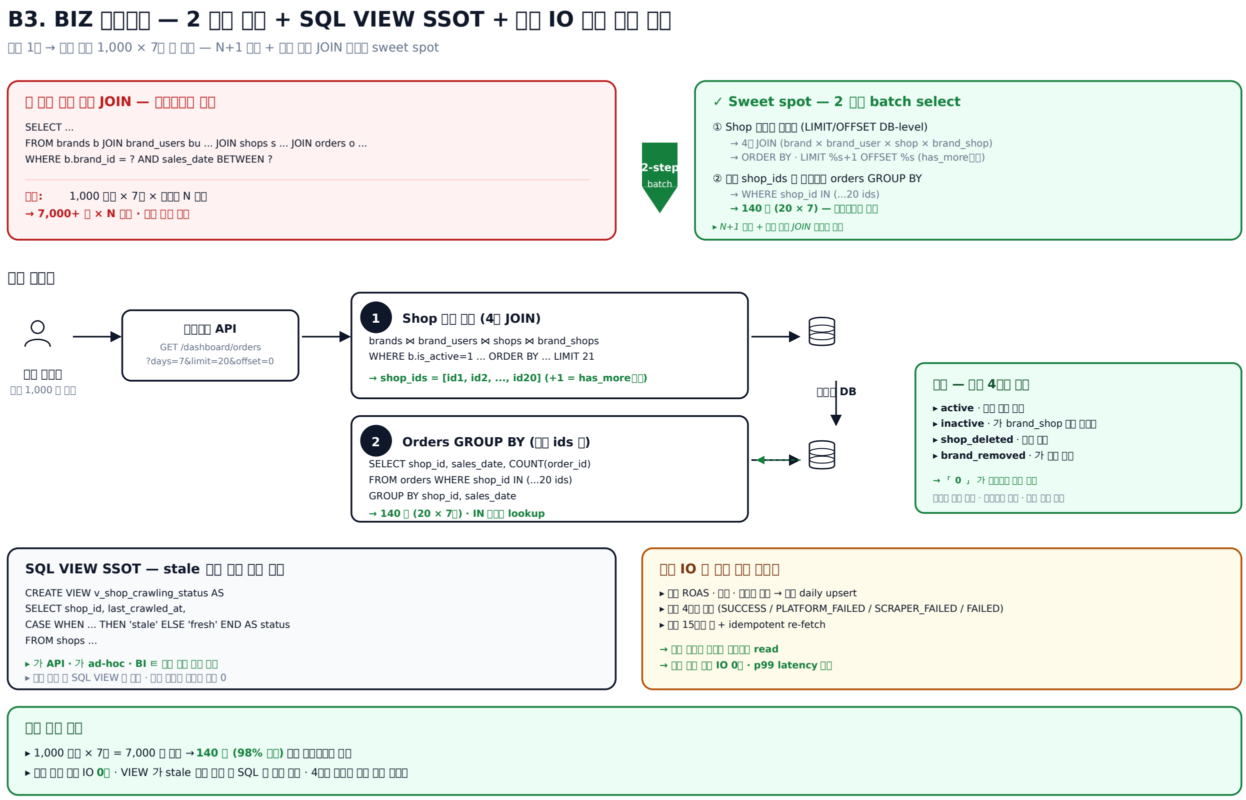
□□ (□□ □□)

- **□ □□ 800□ → 90□ □ (88.75% □□)** — □□ □□□ □□
- **□□ □□□ 70% → 98%** — □□□ □□ □ □□ □□□□
- □□ □□ SPOF □□ (HA □□ □□ □□), □□□ □□□□ □□
- □□ IP □ legacy □□□ □□□□□ □□ **0□**
- CROSSSLOT □□ **0□** — hash tag {pool} □□ □

□□ □□ — 14 phase phased rollout □ phase □ ADR □□ · □□ · □□ · □□ □□ □□□ □□ □□. □□ AI □□ □□ follow-up □□.

Spring □□ — Spring Cloud Gateway + Resilience4j Retry / CircuitBreaker / Bulkhead □□ · WebClient ExchangeFilterFunction □□ proxy resolver □□□ · @ConfigurationProperties + @Validated □ env □□ (□□□ □□□□ [commerce-external-gateway-kt](https://github.com/PreAgile/commerce-external-gateway-kt))

BIZ □□□□ — 2□□ □□ + SQL VIEW + □□ IO □□ □□ □□



BIZ □□□□ 2□□ □□ + SQL VIEW SSOT

□□ — □□□□□ □□ 1□□ □□ 1,000□□□ □ □□□□ □□□.

□ □□□ □□□ □□ □: □□□ □□ 7□ □□ □□, □□ ROAS / □□ / □□, □ □□ 7□, □□□ □□ (stale) □□, □□□ □□.

- □□ JOIN □ □ — brand → … → orders □□ □□ □□□□□ □□ (1,000 × 7 × □□□ N)
- □□ □□ / □□ IO □ □□ □□□ □□ □□ □□ □□
- ORM □□□□□ 4□ □□ + dynamic where + order by + LIMIT □□ □□□□ □□□

□□ — 「N+1 □ □□ □□ JOIN □□□ sweet spot □ 2 □□ batch select」 + 「SQL VIEW □ □□ SSOT 」 + 「 IO □ □□ □□ □□□□ □ 3 □.

- **Step 1** — shop □□□ □□□ (brand × brand_user × shop × brand_shop 4□ □□ · LIMIT %s+1 OFFSET %s has_more □□□□ DB-level pagination)
- **Step 2** — □□ 20 shop_ids □ □□□□ orders GROUP BY → 140 □ (20 × 7) □□□□□ □□
- □□ 4□□ □□ (active / inactive / shop_deleted / brand_removed) — □□ □□□□ □□□ 0」 □□□□
- stale □□ □□□ SQL VIEW (v_shop_crawling_status) □ □□ □□ API · □□□ ad-hoc · BI □□□ □□ □□ □□ (single source of truth)
- □□ / □□□ □□ daily upsert + 4□□ □□ (SUCCESS / PLATFORM_FAILED / SCRAPER_FAILED / FAILED) — □□ □□□ □□□ □□□□ read

□□ (□□ □□)

- 1,000 × 7 = 7,000 □ □□ → **140 □ (98% □□)** □□ □□□□□ □□

- □□ □□ □□ IO **0**□ · p99 latency □□
- VIEW □□ □□ □□ — stale □□ □□ □ SQL □ □□ □□
- □□ 15□□ □ + □□ idempotent re-fetch □ □□ □□□ □□

□□ □□ — 2□□ batch select □□ · VIEW SSOT □□ · 4□□ □□ ADR □□ □□ □□.

Spring □□ — Spring Data JPA + QueryDSL □□ □□ · @Subselect SQL VIEW □□ · @EntityGraph N+1 □□ · Cursor □□□ □□□□□□ (□□□ □□□□ [commerce-comment-platform-be](https://github.com/PreAgile/commerce-comment-platform-be))

□□ □□ □□ — □ □ □□

□ 5□ □□ □□ □□ □□ □□·□□□ □□ □□ (□□□ □□□ □□ / □□□□□□□□):

- **Single-Flight Coordinator** — Hexagonal Architecture (Port-Adapter) + Decorator 5□ □□. □ □ □□ □□ N □□ → □ □□□ □□ □□□, □□□□ □□ □□. □□ □□ 7□□□ spec □□□ □□□. □□□ 80+ □ □□ · 22 new unit spec.
- **SafeCron** □□□□□ — □□ □□□□ cron □□ □□ □□. NestJS DI □□ (setModuleRef) □□□□ □ □ □□□□□□ □□ □ + Slack □□□ + □□□□ □□ □□. 14□ □□.
- □□ □□ □□ — □□ 1,000+ □□ □□ □□. SELECT FOR UPDATE □□□ + □□ □□ □□□□ □□ (@Transactional). □□ □□ / □□ row □□ □□.
- □□□ □□□□ □□□□ □□□□ — 200+ □□□ □□ □□□ □□□ □□ (1 □□ ≈ 1GB+ RAM). □□□□ □□ □□ + □□□□□ ASN □ □□□ □□□ □□ □□□□ **32 □□ / 512 GB RAM** □ □□ **8** □□ □□. Traefik □□ □□□ + 5-step □□□ □□□ □□□□□.
- □□ □□ □□ □□□□□ — □□ → AI → TOCTOU 4 □□□ → API Key/JWT □□ □□ □□. □□□□ □□ □□ + Layered Idempotency □□ □□.
- □□ □□ □□ — 6 □□ □□□ □□ (Shannon □□□□ · zlib □□□ · n-gram · □□□ run) + 가 □ □(Bell + Polarity Amplification + tanh box) + □□·□□□ OR □□.

JVM/Spring □□□ □□□□ — Node.js □□ □□□ Java/Kotlin/Spring □□ □□

5□ □□ Node.js / NestJS □ □□□□ □□ □□□ □□ □ · □□ □□□ · □□□□ □□ · □□ □□□ □□ □□□ Java/Kotlin/Spring □□ □□ □□□ □□□□□ □□ □□□□, □□ □□□ **ADR (Architecture Decision Record — □□ □□□ □ □□ · □□ · □□□ □□ □□□□ □□ □□ □□)** □ □□□ □□□□□□□□.

□□ □□□ NestJS □□□□ □□ □□□, □□ □□□ Java/Kotlin/Spring/JPA □□□□ □□□□ ADR □ □□□□ □□□□□□.

□□□□ **ADR 49□ · □□ 47□ · □□ □□□ 17□** □□ (□□□ □□ — □□ □□□ □□).

[commerce-comment-platform-be](https://github.com/PreAgile/commerce-comment-platform-be) — □□ □□□ (Java / Spring Boot / JPA)

- Stage 0 (raw JDBC) → Stage 5 (DDD / MSA) □□ · Lombok □□ · record / sealed □□ · OSIV=false
- **ADR-006 □□ □□□ 4□□** — idempotency_key □□□ + 4 □□□□ + Redis SETNX 1□ + DB UNIQUE □□ □ □□ + □ 1□ reconciliation 4□ □□□ (Stripe / □□ □□)
- **ADR-004 □□ □** — Redisson watchdog vs SET NX+Lua □□ □ Redisson □□
- **ADR-BE-007 □□□□** — REPEATABLE READ □□ + RC □□ (EXP-06 4 □□□□ phantom □□)

- **ADR-BE-008 □□□□ □□** — □□ □□□ □□ □□□ □□□ □ □□ (EXP-09b 9 □□□□ □ □□ □□)
- **ADR-BE-009 No-Offset Pagination** — Cursor □□□ · row constructor □□ □□ (EXP-07 1M OFFSET 171ms → Cursor)
- □□ □□ — EXP-02 □ 4□ GET_LOCK □□ □□, EXP-03 @Version □□□ silent Lost Update 74□ / OptLockEx 86□ □□, EXP-04 N+1 4-depth 121 prepStmt → JOIN FETCH 1, EXP-13 dirty checking 132x readOnly □□, EXP-14 IDENTITY=saveAll batch □□□ □□

[commerce-batch-orchestrator](https://github.com/PreAgile/commerce-batch-orchestrator) — □□ / Kafka / Outbox

- **ADR-005 Outbox □□ → Debezium CDC □□** — Dual-Write □□ □□
- **ADR-MQ-006 RabbitMQ vs Kafka □□ □□** — RabbitMQ classic queue replay 가 · HOL blocking · Consumer Group □□ □□ □□□ □ Kafka □□ + RabbitMQ □□
- **ADR-MQ-007 Spring Batch Reader 4□ □□** — QuerydslZeroOffsetItemReader □□
- **ADR-MQ-008 Consumer □□ + DLQ + retry topic** — At-least-once + □□ = Effectively-Once
- **ADR-MQ-009 CooperativeStickyAssignor (KIP-429)** — Rebalance stop-the-world □□
- □□ □□ — E-MQ-01 Direct publish □□ → E-MQ-02 Outbox □□ → E-MQ-03 Debezium CDC, E-MQ-04~07 Reader 4□ 100□ row □□, E-MQ-08/09 RabbitMQ ↔ Kafka throughput, E-MQ-10 Eager vs Cooperative Sticky rebalance □□

□□□ *RabbitMQ* □□ □□□□□□ □□□ □ □□□□□ *ADR-MQ-006* □□ □□□ □□ □□□□□.

[commerce-external-gateway-kt](https://github.com/PreAgile/commerce-external-gateway-kt) — □□ □□□□□ (Kotlin / Coroutines / Resilience4j)

- **ADR-002 Coroutines vs Virtual Threads** — Coroutines □□ (□□□ □□□)
- **ADR-007 Circuit Breaker □□□** — 50% / 70% / 90% □□ □ 70% □□ (CB-1/2/3 □□)
- **ADR-SCR-008 Single-Flight Coordinator** — TS □□□□ 7 □□ □□ □□□ Kotlin Coroutines Mutex / Deferred □ □□. SF-2 99.9% coalescing □□
- **ADR-SCR-009 3-Layer □□ □□** — Memory L1 → Redis L2 → □□ □□ L3 (NAVER □□ □□ □□ □□□)
- **ADR-SCR-010 Bulkhead □□** — Semaphore vs FixedThreadPool □□ □□□□ □ □□
- **ADR-SCR-011 Resilience4j □□□□□ □□ □□** — Bulkhead → CB → Retry → TimeLimiter → RateLimiter (Retry □□ CB □□ CB □□ N□ □□□ □□)
- □□ □□ — EXP-WC-01 WebClient .block() EventLoop pinning □□□ 1/10 □□ □□ → □□ □□□ □□

□□ — □□□□□(I-BRICKS) | | □ □□□ 2021.05 ~ 2024.11

□□□ □□□ □□ □□ □□□□ □□·□□ □□□, □□□ □□□□□, □□ □□ □□.

- **EBS □□ □□□ □□□ □□□□□ (Kafka · Apache Nifi · Elasticsearch)** — □□ □□□ □ □□□ Kafka producer / consumer + Nifi flow + Elasticsearch □□□□□ □□. □□ □□□□ INSERT □ □□□ □□ → Kafka □□□ □□□ + Nifi □□ □□□ □□□ 2□ □□□□ □□ □□, □□ □□ □□ **50%** □□. (Kafka □□ □□□ □□ — JVM □□□ □□□□□ commerce-batch-orchestrator □ □ □□□ □□)

- [illegible] [illegible] [illegible] [illegible]·[illegible] **(Elasticsearch + Logstash)** — RDB → ES [illegible] [illegible], [illegible] [illegible] [illegible]·[illegible] [illegible] [illegible] [illegible] [illegible] [illegible]·[illegible] [illegible] [illegible] [illegible] [illegible] [illegible].
- [illegible] [illegible] [illegible] — React / Redux / SCSS [illegible] [illegible] UI, KWCAG 2.1 [illegible] [illegible] [illegible].

[illegible]

F-Lab Java Backend Mentoring | 2024.01 ~ 2024.07

Meta [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible], [illegible] [illegible], [illegible] [illegible] [illegible] [illegible] [illegible], CQRS, [illegible] [illegible] [illegible] [illegible].

[illegible] [illegible] [illegible]